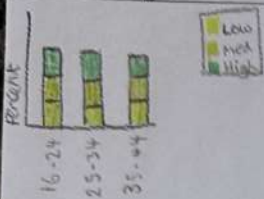


IDEAS

Psychological distress across ages



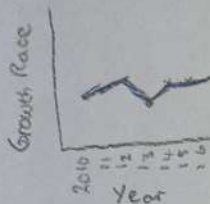
Suicidal thoughts across regions (16-24)



Suicidal thoughts by gender (16-24)



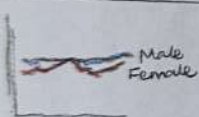
Mental Health spending over time



Mental health disorder by ages



Suicide rates over time



Mental Health Consultations by age and region



Types of disorders (16-24)

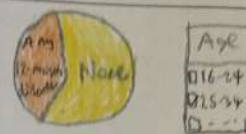


Mental Health disorder by severity



FILTER

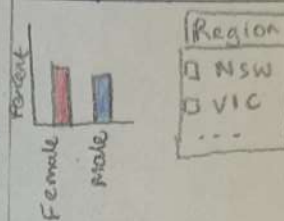
Mental Health disorder by age



(16-24) Suicidal Thoughts



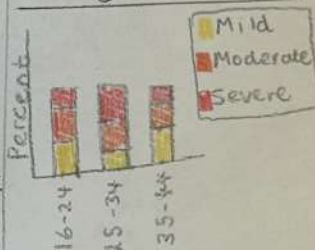
Mental Health consultations by gender and region



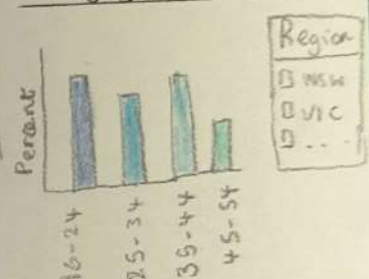
Health spending over time



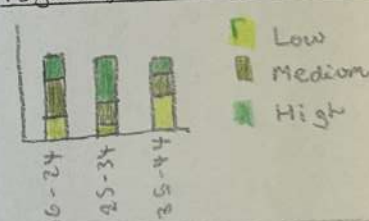
Mental Health disorder by severity



Mental Health Consultations by age and region



Psychological distress across ages



Sheet:1

Author: Esser Ibiann

CATEGORISE

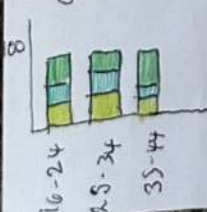
Mental Health Trends

Focus on Youth

Response to mental health

COMBINE & REFINE

Psychological distress



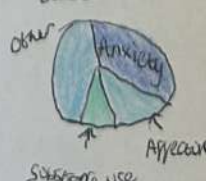
Suicidal thoughts (16-24)



Health Spending over time



Types of mental disorders



QUESTIONS

- 1) Is the scope of the vis realistic?
- 2) Will the message of the visualization be easily followed?
- 3) How can this be implemented in a way that is respectful and sensitive to others feelings?

Layout

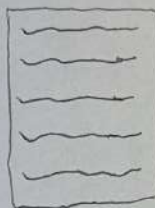
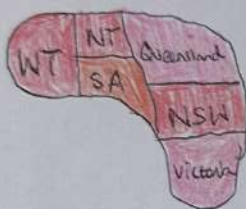
Youth at Risk

Psychological distress across ages

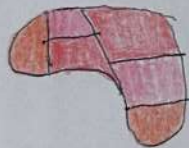


Explanation of Patterns in Stacked bar chart

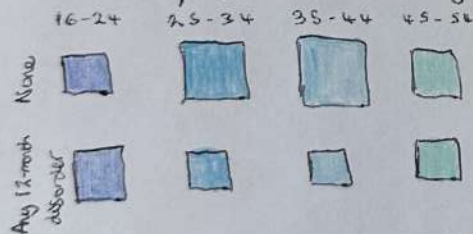
Subsistal thoughts (16-24) by region



Any 12-month disorder by region



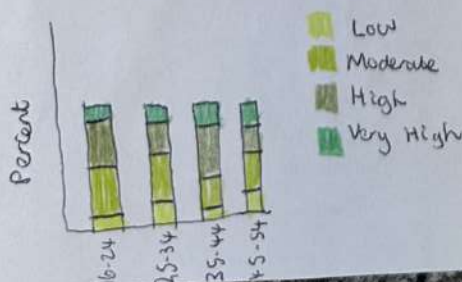
Prevalence of disorder across age groups



Conclusion

Focus

- Exploring if youth have disproportionately high levels of mental health compared to across ages and showing the variation in regions



Title: Youth at Risk

Author: Esther I

Date: 24/09/2025

Sheet: 2

Task: FIT3179

Operations

Stacked bar chart (psychological distress across ages):

- Option to toggle categories (e.g. focus only on low distress)

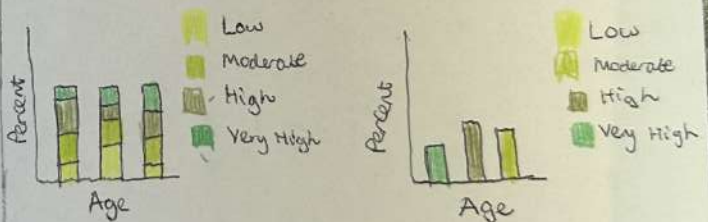
Choropleth Map:

- Click on region directly on map → Shows exact values

Heatmap:

- Hovering over shows breakdown of proportion per each age group

- Hovering over bar gives the exact percentage of distress for the age group



Discussion

Advantages:

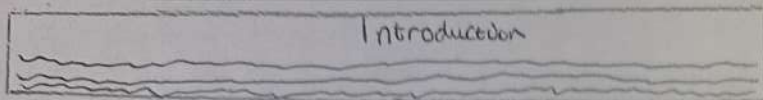
- Having a variety of charts gives a wider and multi-angle view of the different factors e.g. age, region and trends
- Having interactive elements allows users to explore their own region of interest

Disadvantages:

- Doesn't explore the government responses to mental health rates
- Risk of overwhelming users with number of charts if not laid out well

Layout

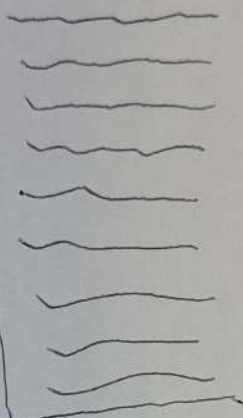
Mental Health Trends within the Youth Population



Types of disorders among youth (18-24)



Explanation of trends among youth



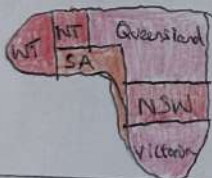
Psychological distress 18-24



Exploring trends across the regions

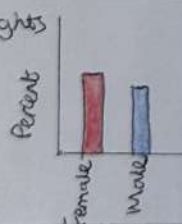
Suicidal thoughts by region

- ☐ Female
- ☐ Male
- ☒ All

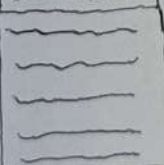


Exploring trends by gender

Suicidal thoughts by gender



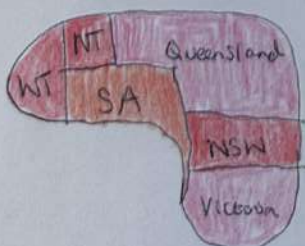
Conclusion



Focus

- To show how mental health rates vary by gender, region and disorder types.

- ☐ Female
- ☐ Male
- ☒ All



Title: Mental Health Trends within the Youth population

Author: Esther I

Date: 24/09/25

Sheet: 3

Task: FIT3179

Operations

- Pie chart → types of disorders (anxiety, affective, substance use)

- Treemap → Psychological distress levels (Low/Moderate/High/Very High)

- Bar chart → Suicidal thoughts by gender (male vs female)

- Map → Suicidal thoughts by region

• Hovering on section in pie chart shows a tooltip with the exact percentage of youth with the disorder

• Hovering on treemap highlights the section and shows exact % proportion

• Users can toggle between female, male and all on the map

- ☒ Female
- ☐ Male
- ☐ All



- ☐ Female
- ☐ Male
- ☒ All



Discussion

Advantages:

- Inclusion of interactivity lets users focus on the aspect they're interested in (gender or region)
- Map combines both gender and region, emphasising the differences among the youth demographics

Disadvantages:

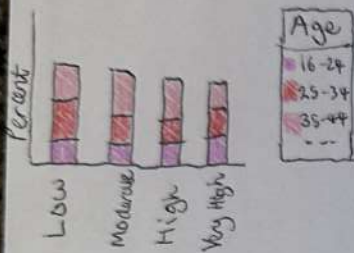
- Some charts may be overlapping or repetitive (suicidal thoughts by region and gender)
- Four different visualisation forms may be overwhelming for users to fully follow along

Layout

Is enough being done regarding mental health among the youth?

Introduction

Mental health disorder severity across different age groups

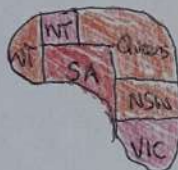


Show difference in
Severity between youth
and other age groups

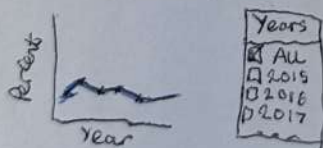
Suicidal thoughts by region (16-24)



At least 1 mental health
consultation by region (16-24)



Mental Health Spending growth over time



Conclusion

Focus

- Response to youth mental health struggles explored through the government's spending and the use of mental health services by youth

Title: Is enough being done regarding mental health among youth?

Author: Esther I

Date: 24/09/25

Sheet: 4

Task: FIT3179

Operations

- Stacked bar chart → Severity of mental health disorder (mild/moderate/high/very high) by age group
- Choropleth map → Suicidal thoughts among youth (16-24) by region
- Choropleth map → Proportion of youth ^{having had} at least 1 mental health consultation by region
- Line chart → Spending on mental growth rate by year
- Clicking on a region in 1 of the maps highlights the region on the other map
- Users can select the years they want to see on the line chart
- Filtering on age groups → Selecting age group shows only its values on the stacked bar chart

Discussion

Advantages:

- Clearly examines and shows a problem (mental health struggles) and the response (spending on mental health services)
- Educational and informative to see where action can take place to improve policies

Disadvantages:

- Important to not misinterpret as direct cause when it is just patterns
- Using data that is both regional (map) and national (spending)

Layout

* Sensitive content warning

Exploring Youth Mental Health Trends in Australia

Introduction

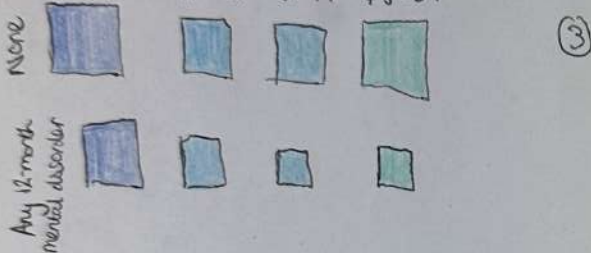
Psychological distress across ages



Types of disorders (16-24)

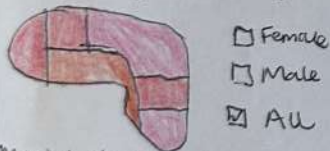


Prevalence of mental health disorders across age groups

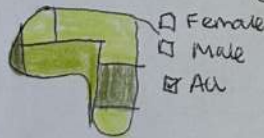


Focussing on youth

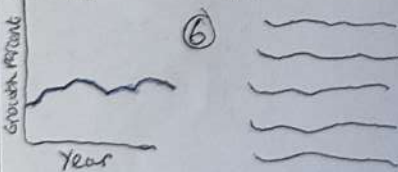
Suicidal thoughts across region (16-24)



At least one Mental health consultation (16-24)



Mental Health Spending growth over time



Conclusion

Title: Exploring Youth Mental Trends in Australia

Author: Esther I

Sheet: 5

Date: 24/09/2025

Task: FIT3179

Operations

① Stacked bar chart: Psychological distress across ages

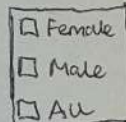
② Pie chart: Types of mental health disorders

③ Heatmap: Prevalence of mental health disorders by age

④ Map: Suicidal thoughts among 16-24 by region

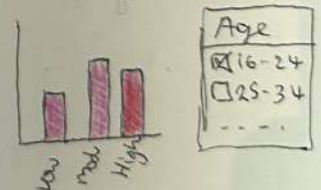
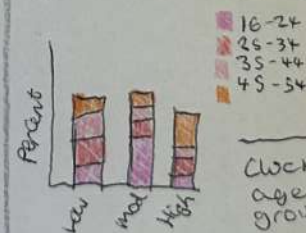
⑤ Map: Mental health consultations among 16-24 by region

⑥ Line chart: Mental health spending growth over time in Australia



Single selection

- Clicking a region in a map highlights the clicked region in the other map



Focus

- To explore patterns in youth mental health by comparing trends in age, gender and region
- To show both the problems (suicidal thoughts, mental health struggles) and responses (mental health consultations, government spending)

Detail

- Vega-Lite for visualisations
- Datasets must be cleaned
- GeoJSON of Australia regions needed for map
- Estimated time: 11/2 weeks
- Data cleaning
- Build charts
- Adding CSS, using HTML, uploading to GitHub
- Important to not remember correlation $\neq$ causation